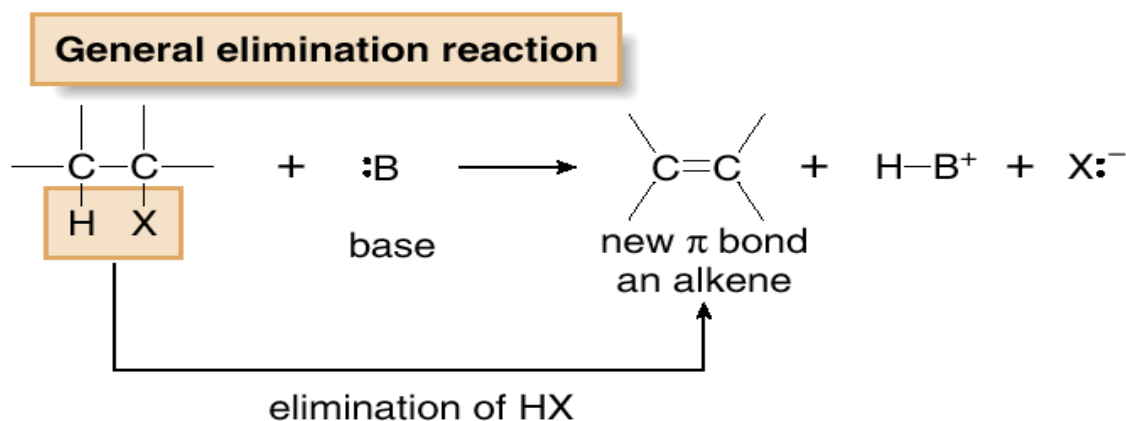


CHE 1203 Organic Chemistry 1

Elimination Reactions (E1 & E2 Reactions)

Elimination Reactions

- Nucleophile molecules such as HO^- , RO^- can act as a nucleophile or a base. When it act as a nucleophile, then substitution reaction takes place. When it act as a base, then elimination reaction takes place.
- Elimination reactions involve the loss of elements from the starting material to **form a new π bond** in the product.
- Alkyl halide + base $\rightarrow$ Alkene
- Alkyl halides undergo elimination reactions with Brønsted–Lowry bases. The elements of HX are lost and an alkene is formed.



Mechanisms of Elimination:

- There are two mechanisms, E2 and E1,
- E1 mechanism—unimolecular elimination
- E2 mechanism—bimolecular elimination
- The E2 and E1 mechanisms differ in the timing of bond cleavage and bond formation, analogous to the S_N2 and S_N1 mechanisms.
- E2 and S_N2 reactions have some features in common, as do E1 and S_N1 reactions.
- Removal of the elements HX is called dehydrohalogenation.

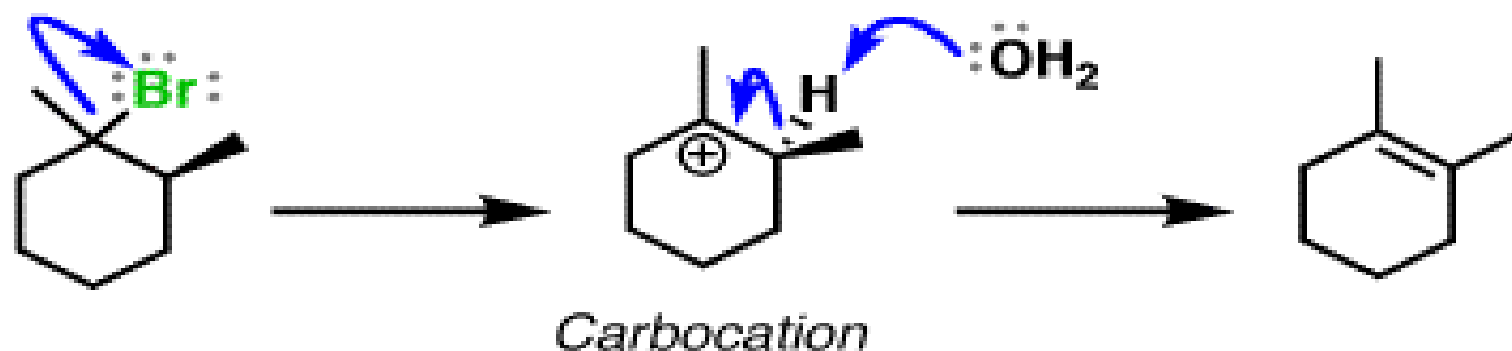
Common Bases Used in Dehydrohalogenation

$\text{Na}^+ \text{ } ^-\text{OH}$	sodium hydroxide
$\text{K}^+ \text{ } ^-\text{OH}$	potassium hydroxide
$\text{Na}^+ \text{ } ^-\text{OCH}_3$	sodium methoxide
$\text{Na}^+ \text{ } ^-\text{OCH}_2\text{CH}_3$	sodium ethoxide
$\text{K}^+ \text{ } ^-\text{OC}(\text{CH}_3)_3$	potassium <i>tert</i> -butoxide

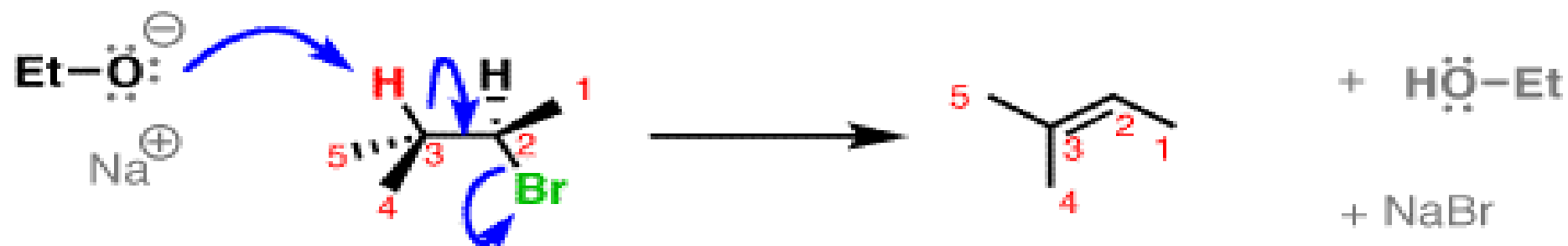
The E₁ Mechanism

Step 1: Loss of leaving group

Step 2: Deprotonation



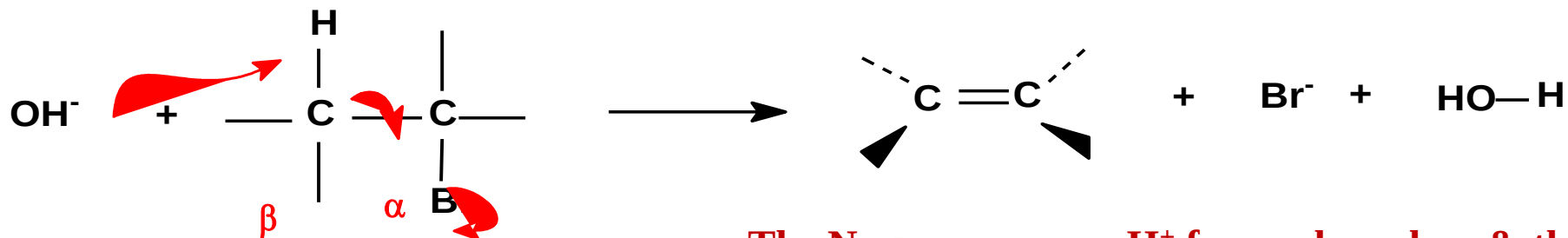
The E₂ Mechanism



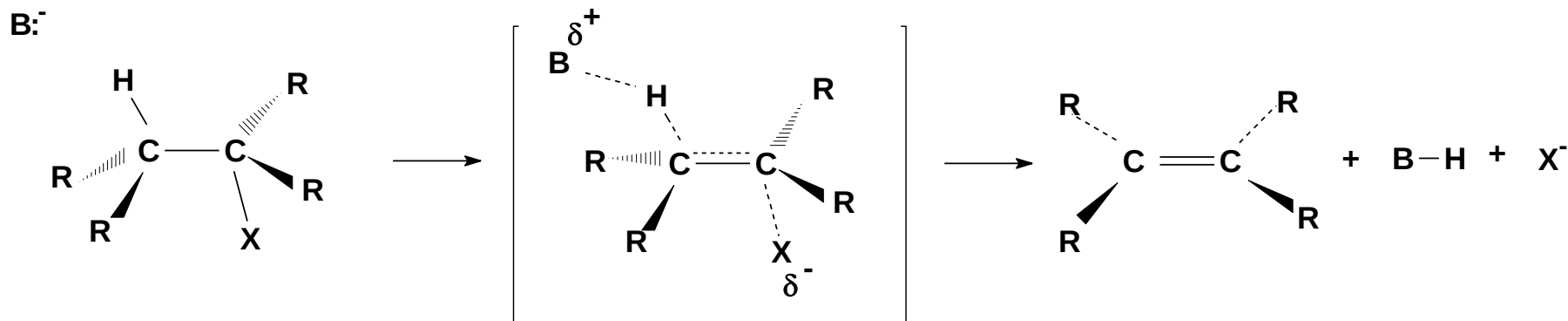
One step! Hydrogen removed must be "anti" to the leaving group (dihedral angle 180°)

E2 Reaction Mechanism

- E2 reactions involve a single step. Base removes a proton from carbon, and simultaneously a halide ion departs and the double bond forms. This is a one-step reaction via a single transition state.
- The E2 reaction is also known as β elimination (remove β H)

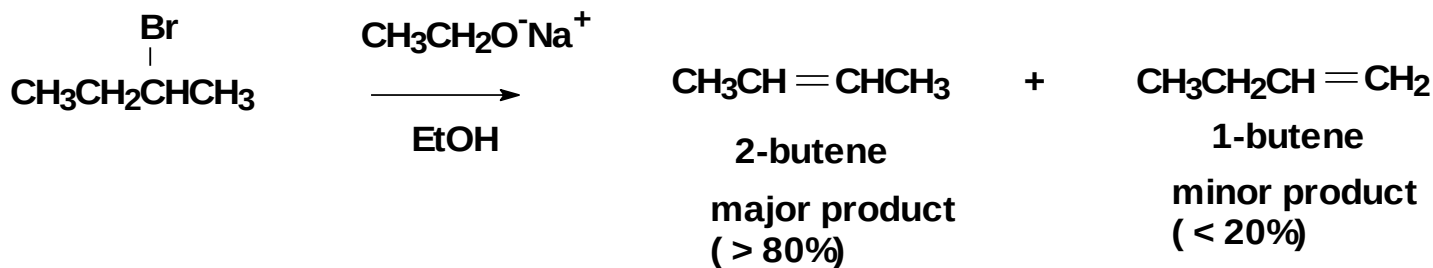


The Nu^- removes an H^+ from a β -carbon & the halogen leaves forming an alkene.



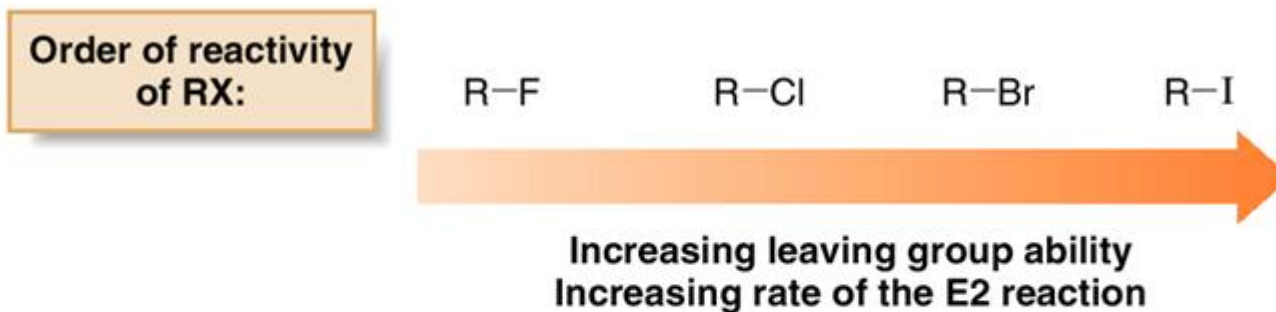
E2 Reaction Mechanism

- In E2 reactions, the Base to H σ bond formation, the C to H σ bond breaking, the C to C π bond formation, and the C to Br σ bond breaking all occur simultaneously at the transition state. No carbocation intermediate forms.
- E2 = Elimination, Bimolecular (2nd order). Rate = $k [\text{RX}] [\text{Nu}^-]$
- Rearrangements. This single step provides no opportunity for rearrangement
- Reactions in which several steps occur simultaneously are called '*concerted*' reactions.
- Zaitsev's Rule:
- Recall that in elimination of HX from alkenes, the more highly substituted (more stable) alkene product predominates.



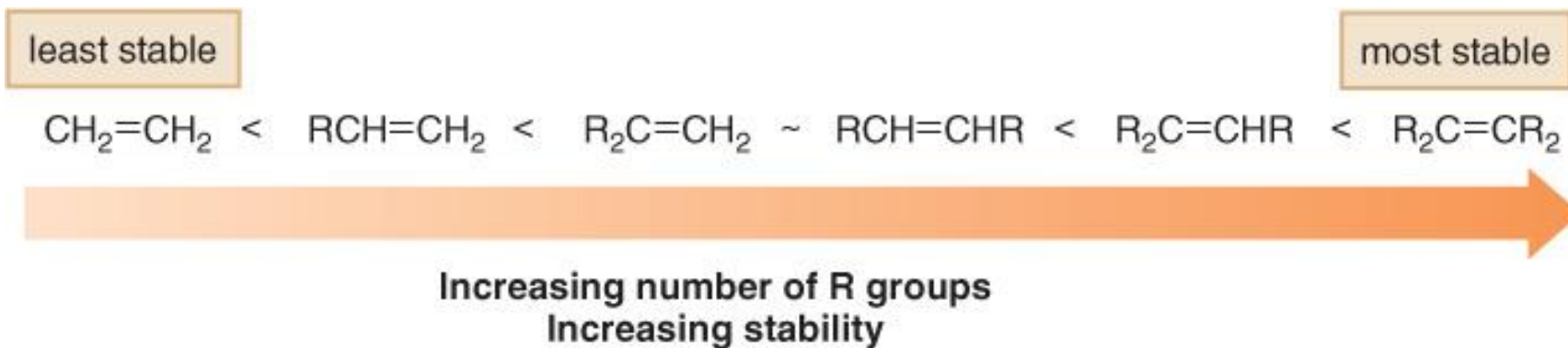
Rate of E2 Depend on

- Ease of breaking the carbon – halogen bond (same as SN reactions)



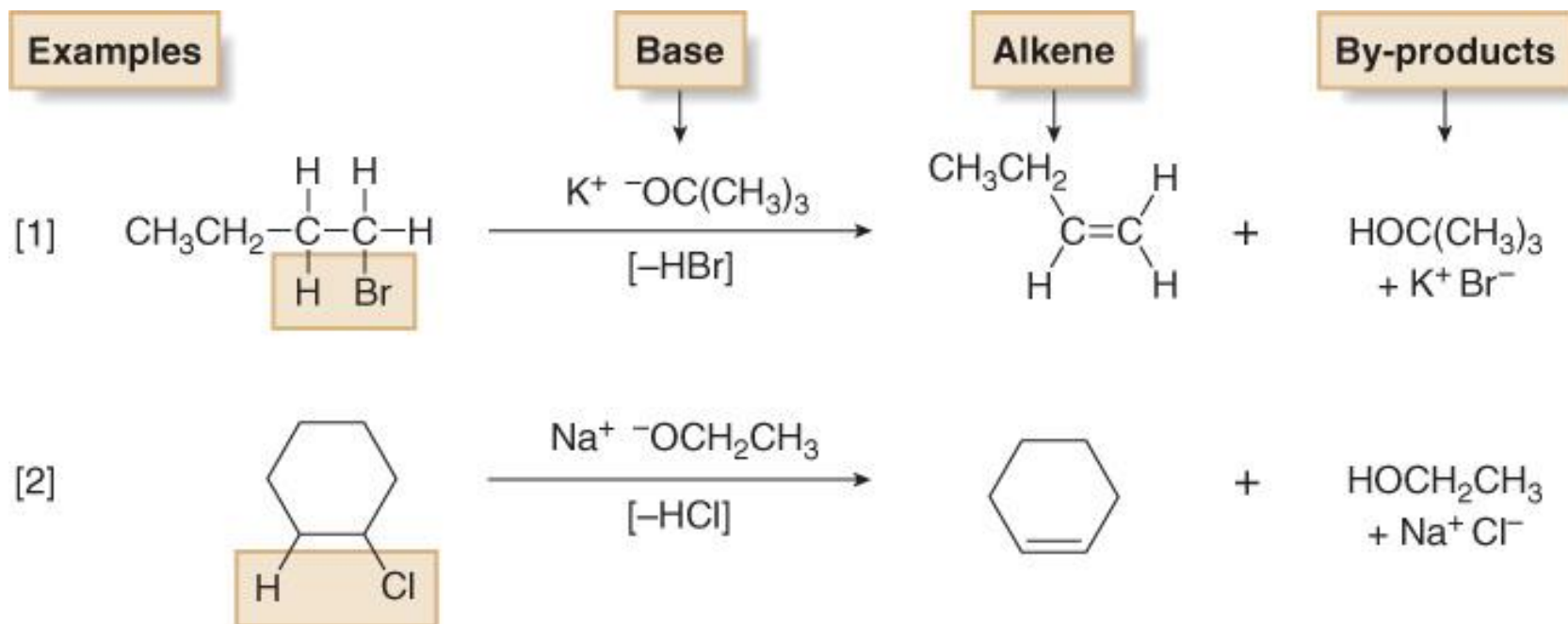
- Because the bond to the leaving group is partially broken in the transition state, the better the leaving group the faster the E2 reaction.

- No of R groups (specially in α and β positions)

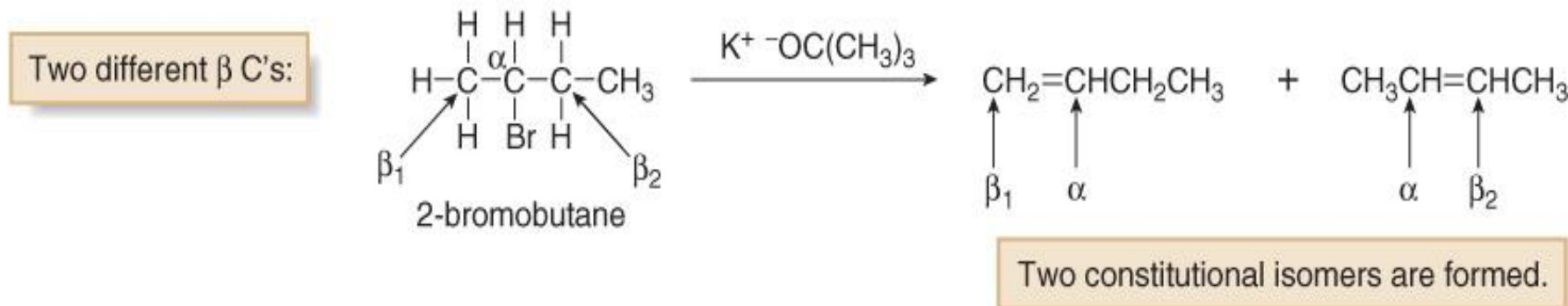
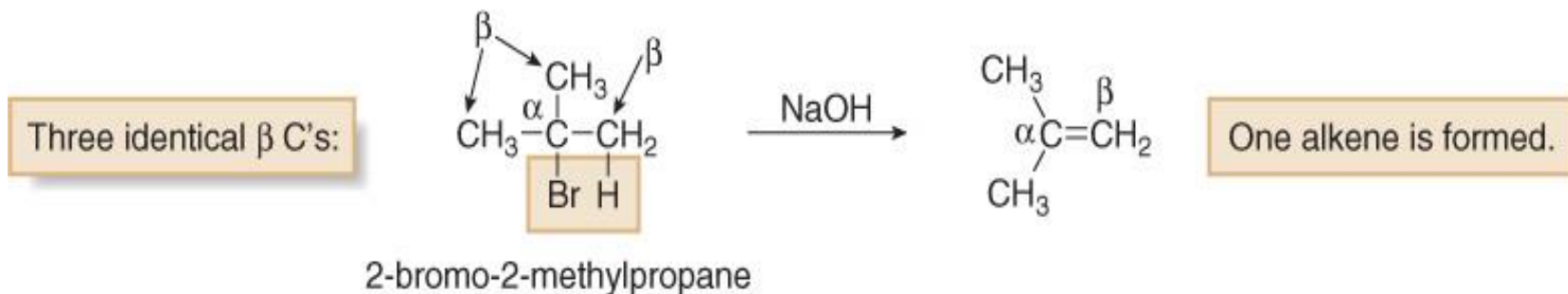


Alkyl Halides and Elimination Reactions

- Equations [1] and [2] illustrate examples of elimination reactions. In both reactions a base removes the elements of an acid, HX, from the organic starting material.



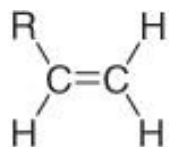
- To draw any product of dehydrohalogenation—Find the α carbon. Identify all β carbons with H atoms. Remove the elements of H and X from the α and β carbons and form a π bond.



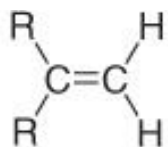
Alkyl Halides and Elimination Reactions

Classification of Alkenes:

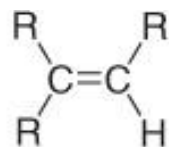
- Alkenes are classified according to the number of carbon atoms bonded to the carbons of the double bond.



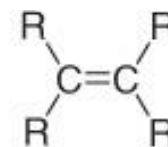
monosubstituted
(one R group)



disubstituted
(two R groups)

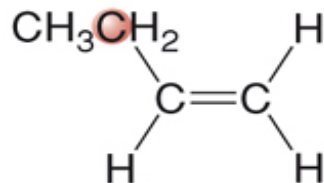


trisubstituted
(three R groups)

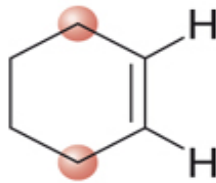


tetrasubstituted
(four R groups)

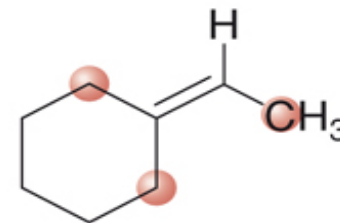
Figure 8.1 Classifying alkenes by the number of R groups bonded to the double bond



monosubstituted



disubstituted

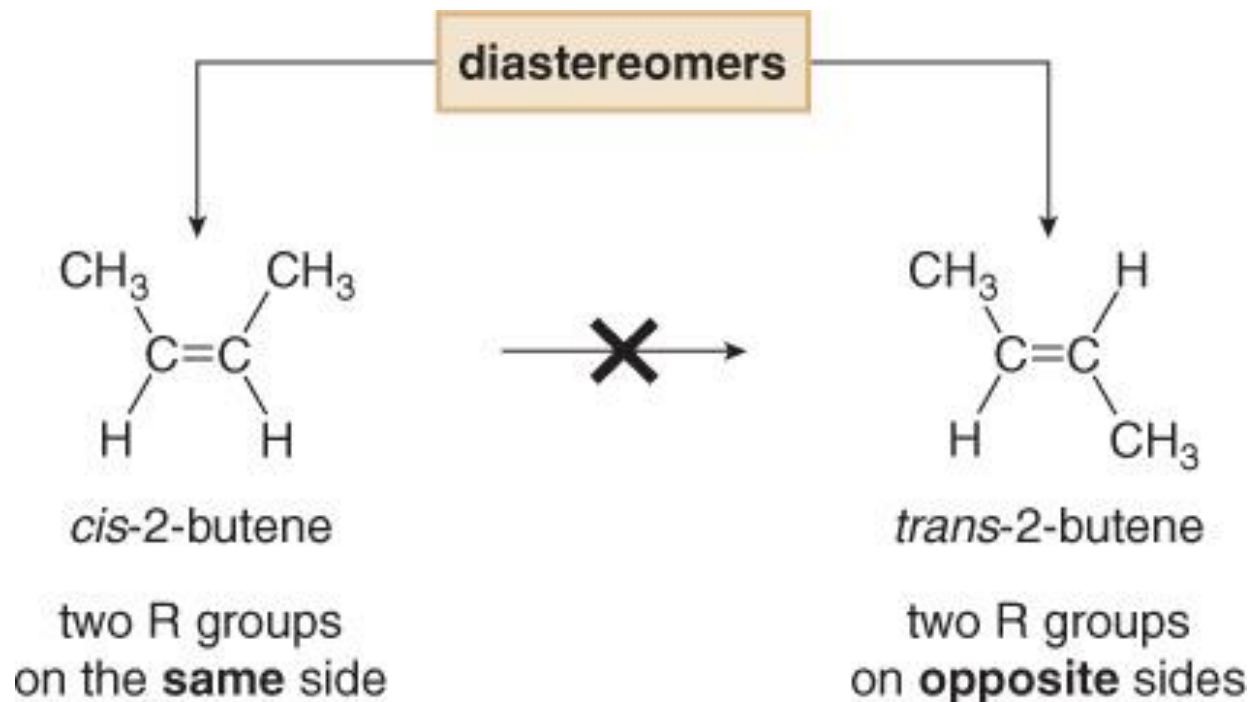


trisubstituted

- Carbon atoms bonded to the double bond are screened in red.

Alkene: The Products of Elimination

- Because of restricted rotation, two stereoisomers of 2-butene are possible. *cis*-2-Butene and *trans*-2-butene are diastereomers, because they are stereoisomers that are not mirror images of each other.

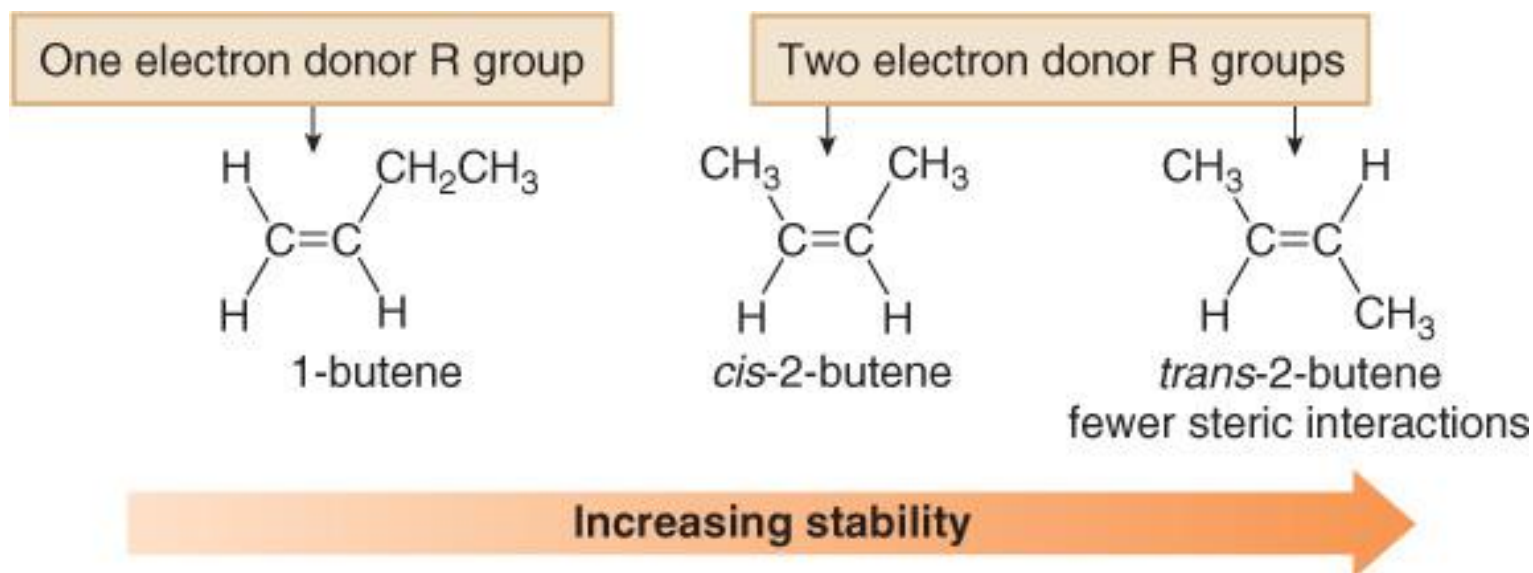


In general, **trans alkenes are more stable than cis**

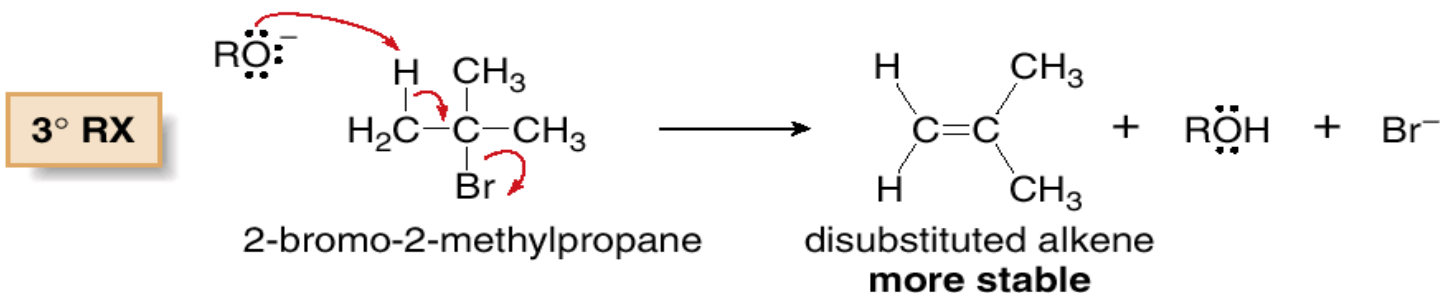
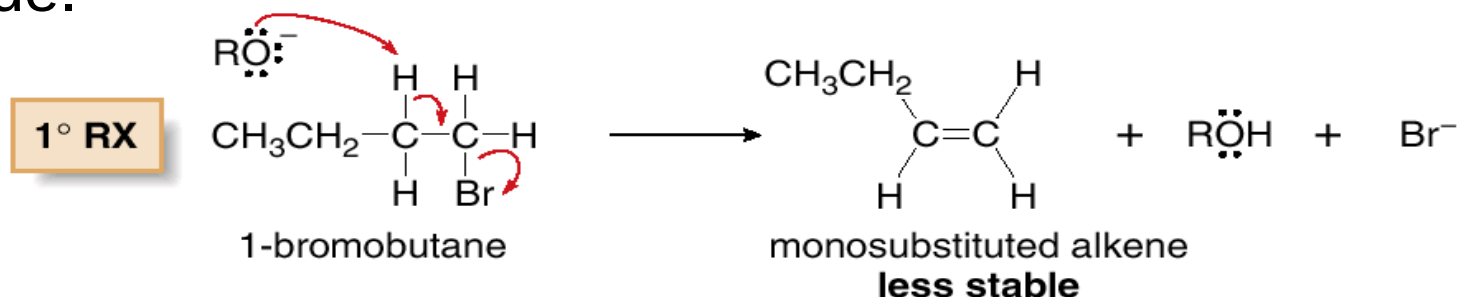
Alkyl Halides and Elimination Reactions

Alkene: The Products of Elimination

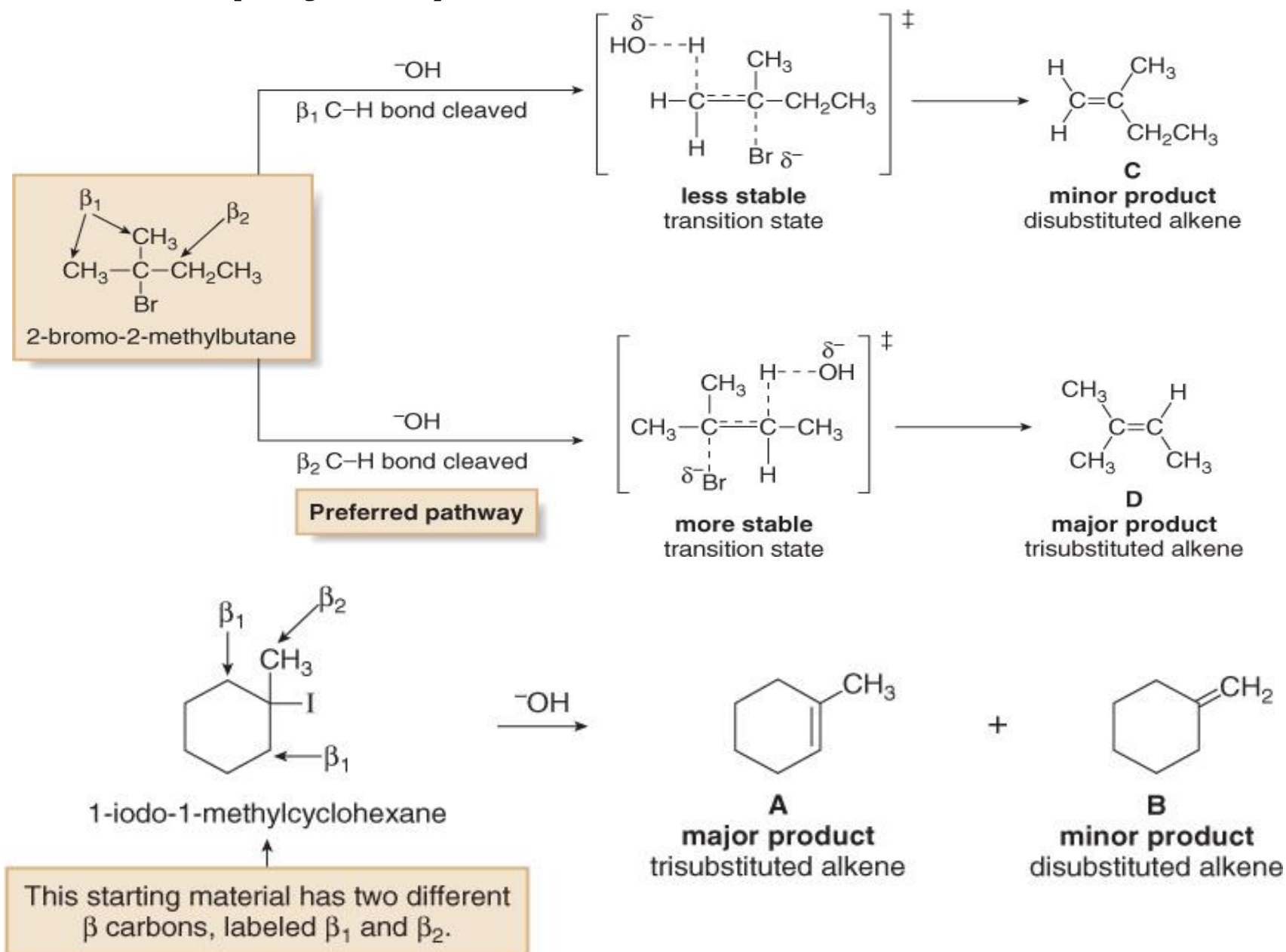
- *trans*-2-Butene (a disubstituted alkene) is more stable than *cis*-2-butene (another disubstituted alkene), but both are more stable than 1-butene (a monosubstituted alkene).



- Increasing the number of R groups on the carbon with the leaving group forms more highly substituted, more stable alkenes in E2 reactions.
- In the reactions below, since the disubstituted alkene is more stable, the 3° alkyl halide reacts faster than the 1° alkyl halide.



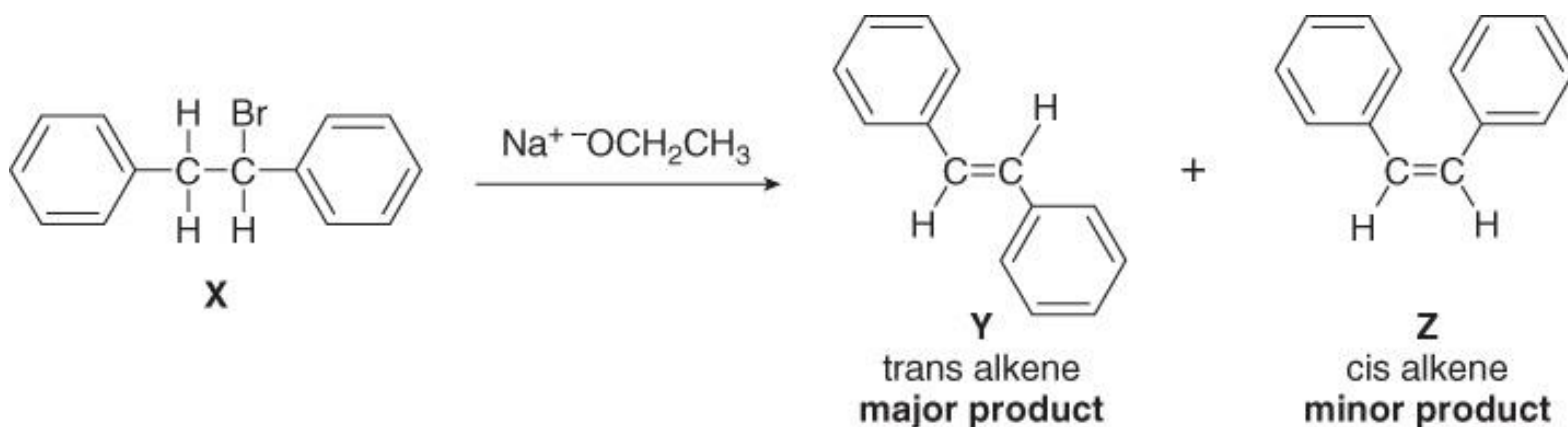
The Zaitsev (Saytzeff) Rule:



Alkyl Halides and Elimination Reactions

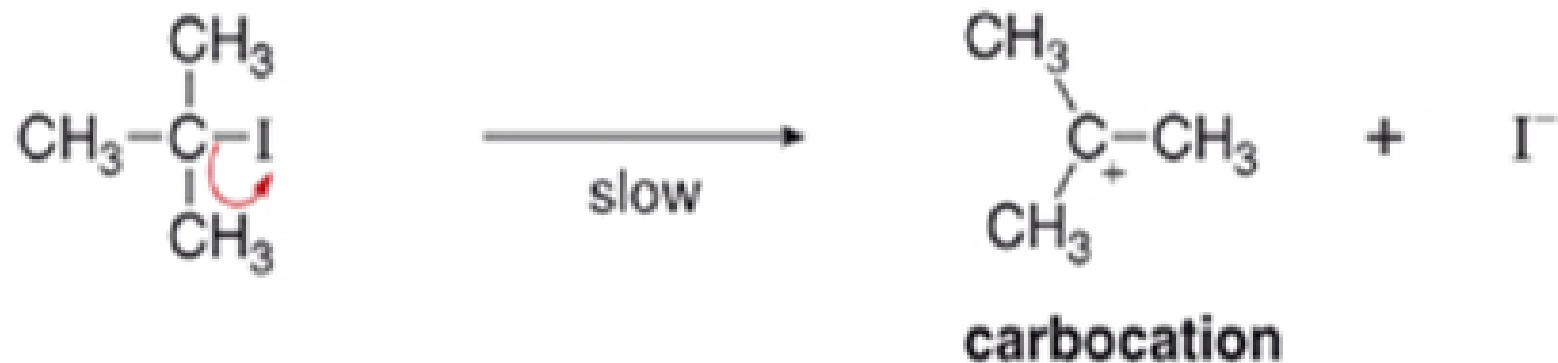
Stereoisomer products:

- When a mixture of stereoisomers is possible from a dehydrohalogenation, the major product is the more stable stereoisomer.
- A reaction is **stereoselective** when it forms predominantly or exclusively one **stereoisomer** when two or more are possible.
- The E2 reaction is stereoselective because one stereoisomer is formed preferentially.

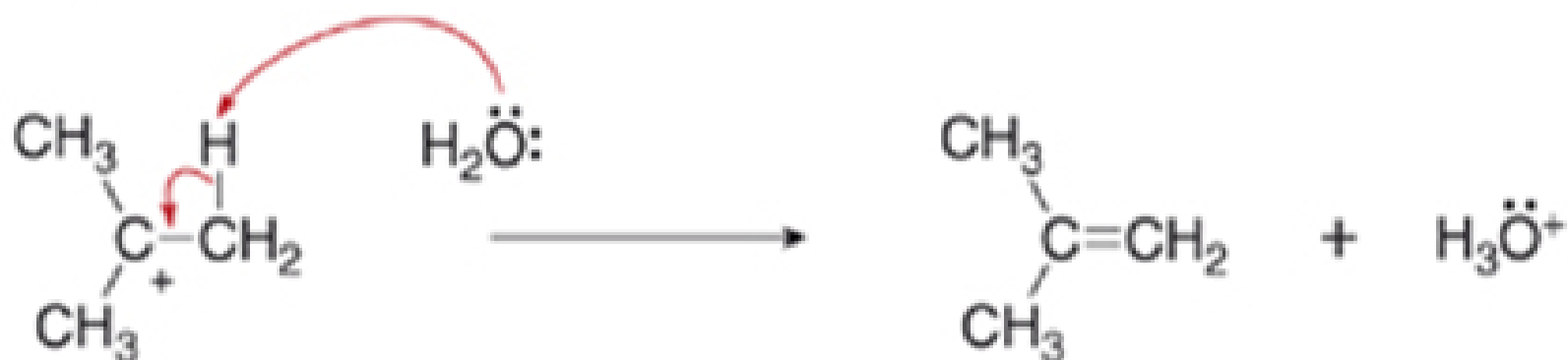


Mechanisms of Elimination: E1

Step [1] The C–I bond is broken.



Step [2] A C–H bond is cleaved and the π bond is formed.



Alkyl Halides and Elimination Reactions

Mechanisms of Elimination: E1

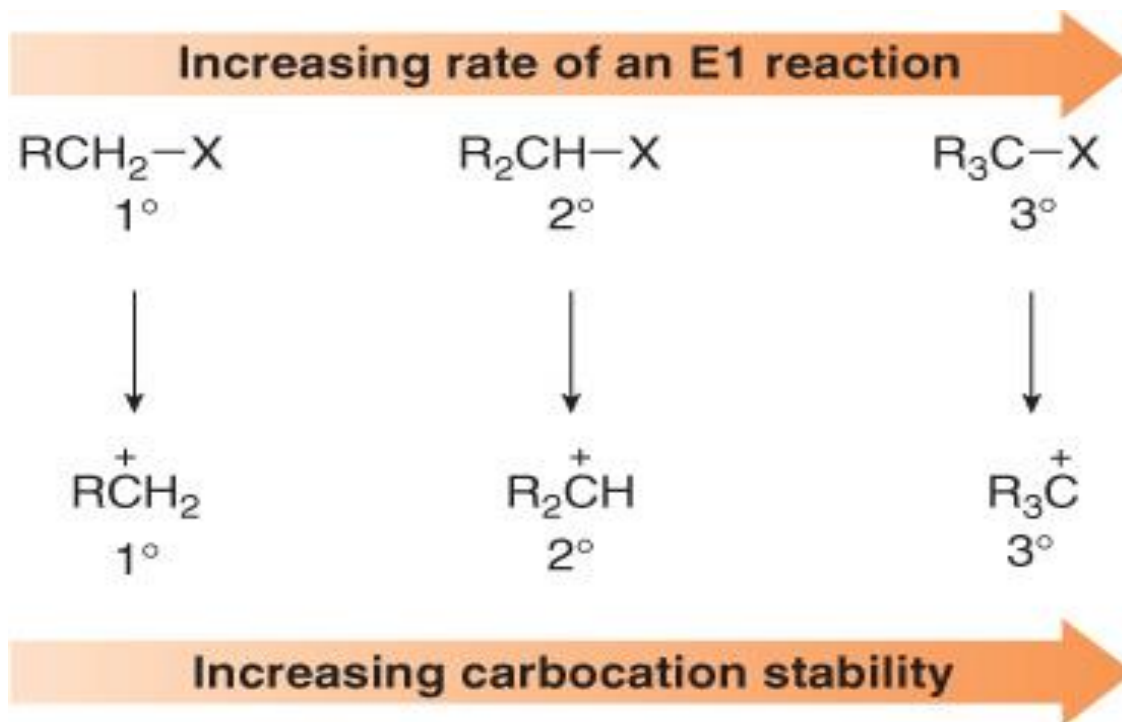
- The dehydrohalogenation of $(\text{CH}_3)_3\text{CCl}$ with H_2O to form $(\text{CH}_3)_2\text{C}=\text{CH}_2$ can be used to illustrate the second general mechanism of elimination, the E1 mechanism.

- An E1 reaction exhibits first-order kinetics:

$$\text{rate} = k[(\text{CH}_3)_3\text{CCl}]$$

- The E1 reaction proceeds via a two-step mechanism: the bond to the leaving group breaks first before the π bond is formed. The slow step is **unimolecular**, involving only the alkyl halide.
- The E1 and E2 mechanisms both involve the same number of bonds broken and formed. The only difference is timing. In an E1, the leaving group comes off before the β proton is removed, and the reaction occurs in two steps. In an E2 reaction, the leaving group comes off as the β proton is removed, and the reaction occurs in one step.

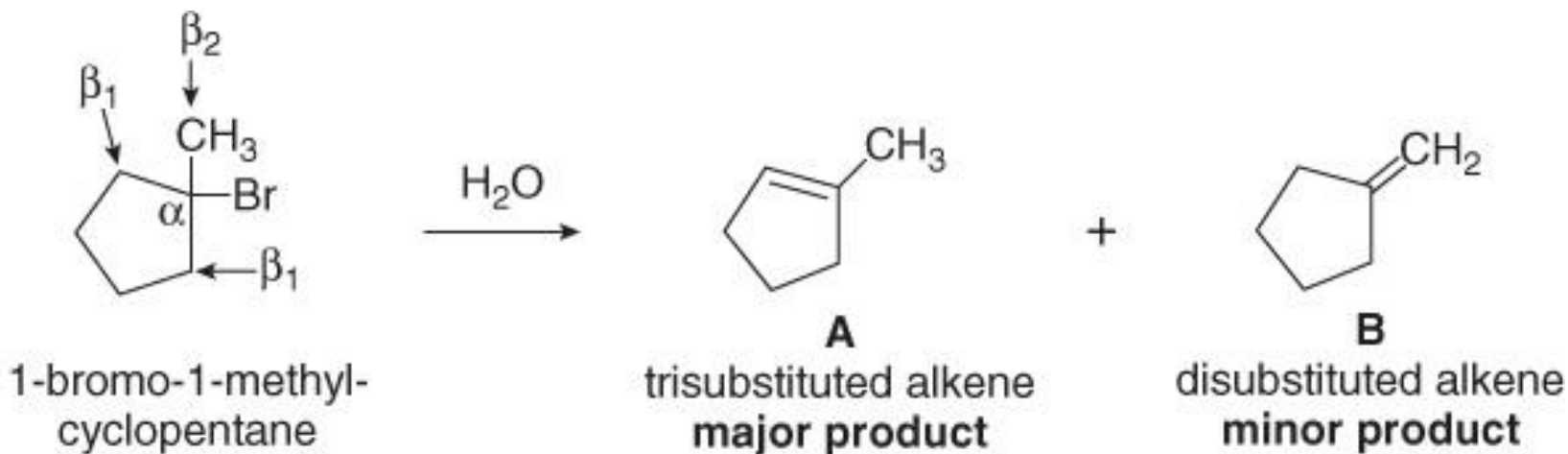
- The rate of an E1 reaction increases as the number of R groups on the carbon with the leaving group increases.



- The strength of the base usually determines whether a reaction follows the E1 or E2 mechanism. Strong bases like OH^- and OR^- favor E2 reactions, whereas weaker bases like H_2O and ROH favor E1 reactions.

Mechanisms of Elimination: E1

- E1 reactions are regioselective, favoring formation of the more substituted, more stable alkene.
- So, Zaitsev's rule applies to E1 as well as E2 reactions.



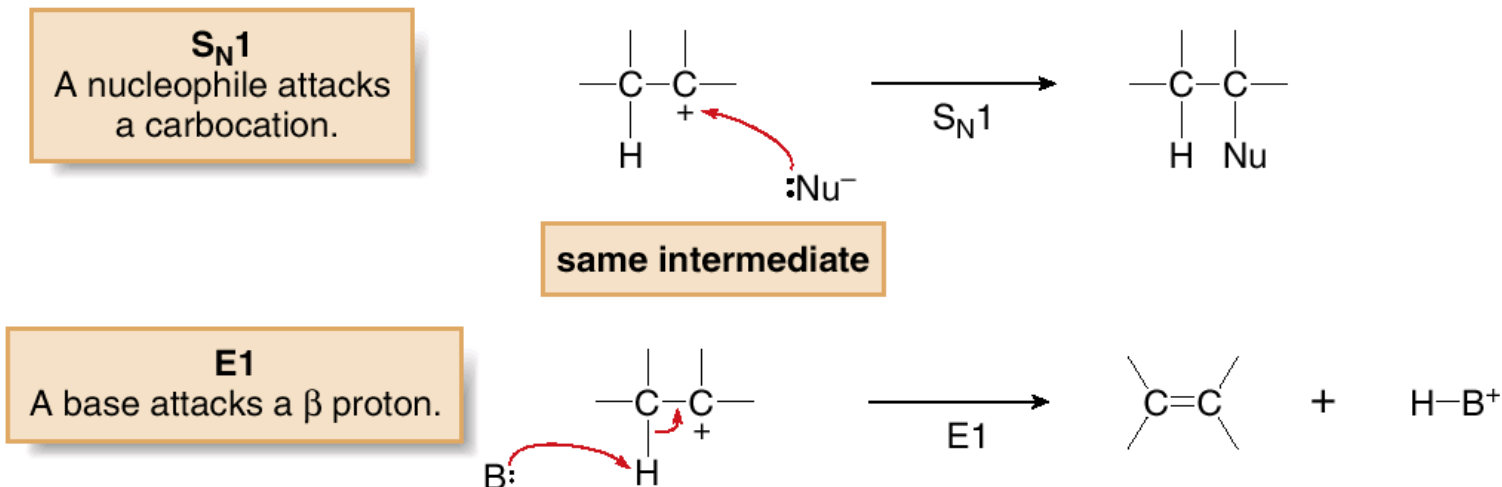
two different β carbons
labeled β_1 and β_2

Mechanisms of Elimination: E1

Characteristic	Result
Kinetics	• First order
Mechanism	• Two steps
Identity of R	• More substituted halides react fastest • Rate: $R_3CX > R_2CHX > RCH_2X$
Base	• Favored by weaker bases such as H_2O and ROH
Leaving group	• A better leaving group makes the reaction faster because the bond to the leaving group is partially broken in the rate-determining step.
Solvent	• Polar protic solvents that solvate the ionic intermediates are needed.

Comparison of S_N1 and E1 Reactions:

- S_N1 and E1 reactions have exactly the same first step—formation of a carbocation. They differ in what happens to the carbocation.



- In an S_N1 reaction, a nucleophile attacks the carbocation, forming a substitution product.
 - In an E1 reaction, a base removes a proton, forming a new π bond.
- Because E1 reactions often occur with a competing S_N1 reaction, E1 reactions of alkyl halides are much less useful than E2 reactions.